Run Length Encoding

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The problem

- Encode runs as count+char, with runs longer than 9 split ("AAAAAAAAAAAAA" → "9A3A").

Intuition

- Count runs of equal characters, but cap each run at nine so the count is always a single digit.

Approach

- 1. Scan with a running count
- 2. Flush when the character changes or the count hits nine
- 3. Write count then character

Java solution

```java
public class RunLengthEncoding {
    public static String runLengthEncoding(String s) {
        StringBuilder sb = new StringBuilder();
        int run = 1;
        for (int i = 1; i <= s.length(); i++) {
            if (i == s.length() || s.charAt(i) != s.charAt(i - 1) || run == 9) {
                sb.append(run).append(s.charAt(i - 1));
                run = 1;
            } else run++;
        }
        return sb.toString();
    }
}
```

Complexity

- Time: O(n)
- Space: O(n)

Recap

- Count runs of equal characters, but cap each run at nine so the count is always a single digit.
- Complexity: O(n)